



Learning from the waist: Aspect-ratio-aware positive sampling for oriented ship detection

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ABSTRACT

Ship detection in earth observation imagery has become a key research focus for maritime resource monitoring and environmental management applications. However, the random orientations, large aspect ratios, and substantial scale variations of ships pose significant challenges for accurate localization and classification. To tackle these challenges, we propose an aspect-ratio-aware one-stage oriented ship detection method with dynamic label assignment. Our approach incorporates three novel strategies: (1) Trigonometric Normalized Coder (TNC), which employs trigonometric functions with normalization to create a concise angle encoder-decoder that effectively resolves angular boundary discontinuity; (2) Ratio-Driven Dynamic Assignment (RDDA), which uniquely utilizes ship aspect ratios to adaptively adjust positive sample selection, particularly incorporating high-quality features from the waist of ship that conventional methods overlook, further enhanced by a two-dimensional Gaussian distribution to measure sample quality; and (3) Attention-Guided Feature Fusion Module (AGFFM), which strengthens multi-scale feature representation by applying spatial and channel attention mechanisms across feature pyramid levels while effectively fusing adjacent layers' features. Extensive experiments on three public datasets (HRSC2016, HRSID, and SSDD+) demonstrate that our approach achieves superior localization and classification performance compared to state-of-the-art one-stage detectors, especially for ships with extreme aspect ratios and varying scales.

1. Introduction

Ship detection using earth observation data has emerged as a vital task in geospatial monitoring and natural resource management, attracting significant research attention in recent years (Gao et al., 2025; Ma et al., 2024; Zhao et al., 2024). This growing interest stems from its crucial applications in environmental protection and maritime governance, including sustainable fisheries surveillance, coastal ecosystem monitoring, illegal fishing detection, and maritime pollution control (Zha et al., 2023; Zhang et al., 2024). The core objective of this geoinformation task lies in accurately distinguishing ships from complex marine backgrounds and determining their precise spatial coordinates through advanced remote sensing image processing and interpretation techniques.

Early ship detection efforts primarily migrate from remote sensing object detection models, leading to the emergence of numerous horizontal bounding box (HBB)-based ship detection models (He et al., 2021). However, HBBs are not the most suitable representation for ship detection tasks. Specifically, ships often appear in densely arranged formations, where the use of HBB can result in a high Intersection over Union

(IoU) between bounding boxes, which adversely affects the subsequent non-maximum suppression (NMS) process, ultimately leading to a decline in detection accuracy (Zhang et al., 2024). Moreover, the high aspect ratio characteristic of ships exacerbates this issue (Tang et al., 2025). Additionally, for ships with distinct directional orientations, the use of HBB fails to capture directional information, resulting in the loss of crucial motion-related details (Wang et al., 2025). Consequently, the use of oriented bounding box (OBB) in oriented ship detection has become a more prevalent research direction in recent studies (Li et al., 2025; Zhan et al., 2024).

Oriented ship detection can be categorized into one-stage and two-stage detectors based on the number of model stages. In the category of one-stage detectors, methods such as R3Det (Yang et al., 2021a), employ feature refinement modules to align rotated features. Similarly, S²ANet (Han et al., 2021a) introduces the Aligned Convolution, which aligns features based on corresponding anchor boxes, enabling rapid ship detection. Oriented RepPoints (Li et al., 2022a) generates a set of key-points from the image and predicts relevant parameters to produce rotated bounding boxes, eliminating the complex hyperparameter tuning

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issues inherent in anchor-based designs. CHPDet (Zhang et al., 2021a) employs an oriented rotated bounding box with a head point to represent ship targets, determining the object's orientation by locating both the head point and the center point. However, this method increases annotation costs. DCNDarknet25 (Su et al., 2022) integrates angle-free regression-based rotated detection capabilities into the YOLO (Redmon et al., 2016) detection algorithm, enabling rapid ship detection. MidNet (Liang et al., 2023) designs an anchor-and-angle-free detector, leveraging symmetric deformable convolutions to localize key points of ships. SRTNet (Pan et al., 2024) employs a ship saliency enhancement module to extract contour details and learn geometric ship features, improving ship prominence in cluttered environments.

Regarding two-stage detectors, RoI Transformer (Ding et al., 2019) generates Horizontal Region of Interest (HROI) from horizontal anchor boxes and then employs a learnable module to convert these horizontal proposals into Rotated Region of Interest (RRoI), circumventing the need for anchor box selection. Meanwhile, Oriented R-CNN (Xie et al., 2021) achieves higher accuracy by generating HROI and predicting the positions of points on each edge of the HROI to produce RRoI. P2RNet (Chen et al., 2024) employs a two-stage detection framework, where the first stage generates region proposals through keypoint heatmap analysis and the second stage achieves fast and accurate ship detection through an improved YOLO network. NACAD (Yuan et al., 2023) utilizes the Position Refined Module to refine the HROI obtained from the region proposal network, thereby achieving small object detection in remote sensing images. Furthermore, advancements in DETR (Carion et al., 2020), such as AO2-DETR (Dai et al., 2022), explore the application of attention mechanisms to address the technical difficulties in rotated object detection for remote sensing imagery, showcasing considerable promise.

However, in the field of oriented ship detection, several critical challenges remain unresolved. Firstly, the introduction of angular parameters to represent orientation in oriented ship detection leads to discontinuity at angular boundaries due to the periodic nature of angles. Notably, as the angle approaches the predefined range boundary, minor angular deviations induce substantial loss variations due to the discontinuity at the boundary. As shown in Fig. 1(a), two highly similar OBBs may exhibit a strong angular loss, which hinders stable regression of angles near the boundary. Secondly, ships typically exhibit a high aspect ratio, and when extracting features from such high-aspect-ratio objects, the receptive field often struggles to align with the object, resulting in a scarcity of high-quality features. Specifically, when the receptive field of detector becomes too large, the features may include significant background noise, whereas a smaller receptive field risks overlooking features along the long axis, as illustrated in Fig. 1(b). Thirdly, the size of ships can vary significantly due to differences in distance, resolution, and ship type, posing a challenge for detection models to accurately identify large, medium, and small-scale ships simultaneously, as illustrated in Fig. 1(c).

To systematically address these three challenges, this paper proposes an anchor-free, one-stage model based on aspect-ratio-aware dynamic label assignment with three corresponding modules: the Trigonometric Normalized Coder (TNC), the Ratio-Driven Dynamic Assignment (RDDA), and the Attention-Guided Feature Fusion Module (AGFFM). Firstly, we introduce TNC by transforming angular values into a continuous space through trigonometric functions with normalization, elegantly solving boundary discontinuity with a concise mathematical formulation, which differs from previous approaches (Yu & Da, 2023). Secondly, RDDA is designed to leverage ship aspect ratios to intelligently expand the positive sampling region, for increasing the quantity of high-quality features. Specifically, it captures valuable features from the waist of ship-critical areas rich in discriminative information that conventional sampling strategies overlook. Thirdly, we propose AGFFM, which integrates channel and spatial attention mechanisms within the feature pyramid architecture, enabling content-adaptive feature enhancement

and effective multi-scale feature fusion. Our main contributions can be described as follows.

1) We propose TNC, a concise trigonometric encoding mechanism that effectively resolves the boundary discontinuity problem in oriented detection. This approach projects angular space into a higher-dimensional representation, creating smooth transitions across boundaries with a simpler formulation than existing methods, significantly improving the stability and accuracy of angle regression.

2) We design RDDA, an aspect-ratio-aware dynamic assignment strategy that addresses the feature scarcity issue in high aspect ratio ships. By intelligently expanding positive sample selection to the waist of ship and using Gaussian distribution to quantify sample quality, RDDA selects more representative high-quality positive samples, substantially enhancing detection performance.

3) We develop AGFFM, a dual-attention feature fusion module for multi-scale ship detection. By integrating channel and spatial attention mechanisms within the feature pyramid to adaptively enhance and deeply fuse multi-level features, this module enables effective detection across ships of varying scales.

The rest of this article is organized as follows. In Section 2, we briefly review the related work in remote sensing object detection. In Section 3, we detail the implementation details of our proposed method. In Section 4, we present experiments on several remote sensing ship datasets and provide a detailed analysis. Finally, we conclude this article in Section 5.

2. Related work

This section briefly reviews the research landscape pertinent to our method. We first trace the lineage of ship detection in remote sensing images, starting with its foundation in generic horizontal object detection in Section 2.1. We then describe the evolution towards oriented object detection in Section 2.2, which addresses the limitations of horizontal boxes for oriented objects. Finally, we focus specifically on advanced techniques developed for ship detection in remote sensing images in Section 2.3. To provide a clear overview, Table 1 summarizes the main ideas and key limitations of these three related research threads.

2.1. Horizontal object detection

Early remote sensing object detection primarily utilized horizontal object detectors adapted from natural image methods. Classical two-stage detectors such as R-CNN (Girshick et al., 2014) and Faster R-CNN (Ren et al., 2015) generate region proposals using anchor boxes and a Region Proposal Network, with subsequent networks refining these proposals for final predictions. In contrast, single-stage detectors including the YOLO (Redmon et al., 2016) series and SSD (Liu et al., 2016) bypass the region proposal network to directly regress and classify anchor boxes, achieving faster processing speeds though with slightly reduced accuracy. Anchor-free approaches like FCOS (Tian et al., 2019) and CenterNet (Zhou et al., 2019) further discard the anchor box strategy, reducing the associated hyperparameters and thereby making the model configuration simpler and more flexible. In recent years, image processing algorithms based on ViT (Dosovitskiy et al., 2020) have gained prominence due to the attention mechanism, leading to the emergence of detection algorithms such as DETR (Carion et al., 2020) and Deformable DETR (Zhu et al., 2020), which achieve commendable results in object detection.

However, when detecting targets like ships in remote sensing, such horizontal detection methods lose the target's orientation information and include partial background content in the bounding boxes, thereby reducing localization accuracy. Moreover, the frequent dense distribution of ships in remote sensing imagery leads to very high IoU between the horizontal bounding boxes of different vessels. This high-IoU prob-

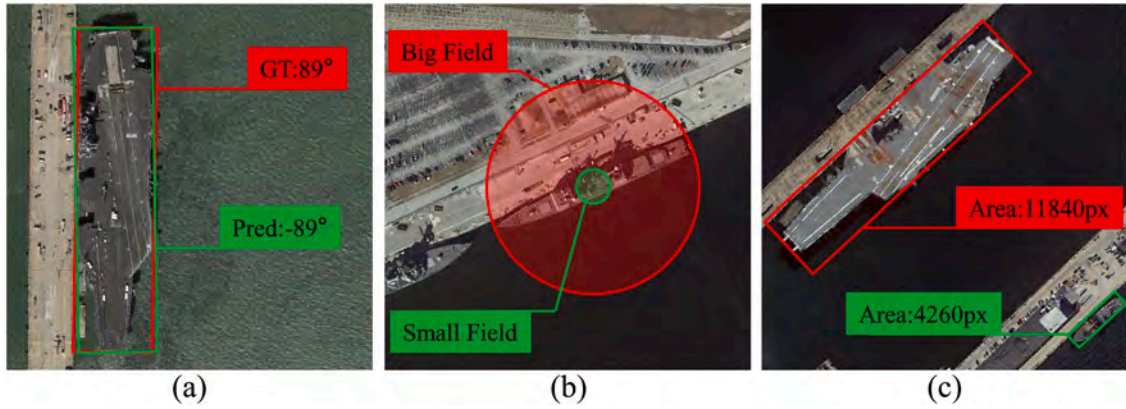


Fig. 1. Challenges in remote sensing oriented ship detection. (a) Angle boundary discontinuity problem. (b) Large aspect ratio problem. (c) Multi-scale objects problem.

Table 1
Summary of related work and associated limitations.

Category	Core Content	Limitations
Horizontal Object Detection	Detects objects using horizontal bounding boxes.	• Loss of Orientation: Ignores target rotation information. • Background Noise: Bounding boxes contain excessive background pixels. • Dense Cluster Failure: High IoU in dense scenes causes NMS to mistakenly eliminate valid targets.
Oriented Object Detection	Detects objects using oriented bounding boxes.	• Aspect Ratio Mismatch: Struggles with slender, high-aspect-ratio ships. • Sample Imbalance: Preset anchors fail to match extreme shapes, leading to insufficient positive samples. • Feature Misalignment: Inadequate receptive fields cause blurred boundaries and low confidence.
Ship Detection in Remote Sensing	Specialized methods for detecting ships in remote sensing imagery.	• Anchor Reliance: Mostly depends on Anchor-based frameworks requiring tedious hyperparameter tuning. • Anchor-Free Gap: Efficient, flexible Anchor-free frameworks remain under-explored.

lem can cause the NMS algorithm to mistakenly eliminate valid detection boxes, consequently resulting in missed detections.

2.2. Oriented object detection

Since horizontal object detection frameworks lack explicit angle parameters, their performance in remote sensing applications is suboptimal. Consequently, a significant body of work has been dedicated to integrating angle parameters into horizontal detectors, adapting them for the domain of oriented object detection. RRPN (Ma et al., 2018) utilizes anchor boxes for region proposal generation, but this approach inherently introduces computational redundancy, ultimately reducing detection speed. Moreover, some works have improved by adjusting the paradigm of rotated box regression. CSL encoding (Yang & Yan, 2020) reformulates angle prediction as a classification problem instead of regression, effectively circumventing the boundary discontinuity issues inherent in direct angle regression. Similarly, PSC encoding (Yu & Da, 2023) designs a differentiable anchor encoder, ensuring that the encoding angle becomes smooth and continuous. On the other hand, ACM encoding (Xu et al., 2024) creates an angle encoder based on Euler's formula. Furthermore, Yuan et al. (2022) redefine a circumscribed rotation box, avoiding a sudden increase in losses during the training process. Zhao and Li (2024) utilized the global and local receptive fields of target to extract spatial contextual features of rotation-dense objects for target classification and localization. Additionally, some works leverage the ideas of DETR, such as ODETR (Ma et al., 2021). By leveraging attention mechanisms, it effectively addresses challenges in remote sensing rotated object detection, showing promising results.

Although existing oriented object detection methods have improved detection accuracy by incorporating angle parameters, they exhibit significant limitations when detecting ships with high aspect ratio. The core issue lies in the mismatch between the detection mechanism and the slender morphology of ships with extremely high aspect ratios. For anchor-based frameworks, the preset anchor boxes struggle to adapt to the extreme length-to-width ratios of such ships, resulting in insufficient positive sample matching and an increased probability of missed detections. Even for anchor-free methods, general feature extraction and label assignment strategies often fail to adequately capture discriminative information along the ship's longitudinal axis. This leads to incomplete feature utilization and receptive field mismatch, which in turn causes inaccurate localization, blurred boundary regression, and low detection confidence for slender ships.

2.3. Ship detection in remote sensing images

In remote sensing object detection, ship targets exhibit distinct characteristics compared to other objects: they generally have a high aspect ratio, and ships near the shore often appear in dense clusters with significant variations in scale. These features present unique challenges for ship detection. Li et al. (2020) enhance the traditional region of interest pooling by developing a shape-adaptive pooling method specifically for ships, which is effective in capturing the spatial distribution characteristics of ships with varying aspect ratios. AF-OSD (Hua et al., 2023) introduces a novel multi-scale dense point sampling strategy to address the positive sample imbalance across varying ship sizes. Guo et al. (2023) propose a shape-aware rotated ROI alignment technique, which effectively mitigates misalignment issues during feature extraction while re-

ducing the number of parameters. Zhan et al. (2024) introduce the multi-scale object enhancement module and an improved multi-task loss function, significantly enhancing the accuracy and fine-grained classification performance of ship detection by suppressing background interference. Wan et al. (2025) propose a pruning framework for ship detectors based on the maximum entropy principle and multitask reinforcement learning, which substantially accelerate inference speed while maintaining near-original detection accuracy. Su et al. (2022) utilized Deformable Convolutional Networks (DCN) to capture irregular ship features, a method whose sampling offsets are entirely network-predicted and which consumes more memory than standard convolution. OSCD-Net (Zhang et al., 2022) pools ship features into multiple feature regions of varying scales to accommodate the aspect ratio of ships, thereby uniformly extracting their characteristics. Li et al. (2022b) proposed a dynamic soft label assignment method to improve anchor-based detectors, which dynamically adjusts the IoU threshold based on the aspect ratios of ships. This ensures that even ships with extreme aspect ratios can retain sufficient positive samples during training. Chen et al. (2019) tackled the sample imbalance between sparse keypoints and abundant background pixels by increasing the weight of keypoint-proximal pixels in the loss function. This strategy ensured the model focused more on critical regions.

To sum up, existing remote sensing ship detection methods have advanced in addressing single challenges such as aspect ratio adaptation, multi-scale feature enhancement or model efficiency optimization. But they lack a comprehensive solution to simultaneously tackle angle boundary discontinuity, dynamic high-quality positive sample selection for high-aspect-ratio ships and adaptive multi-scale feature fusion. Furthermore, most of these methods rely on Anchor-based frameworks, which demand tedious anchor hyperparameter tuning and show poor adaptability to diverse ship shapes and complex backgrounds. In contrast, Anchor-free methods—characterized by greater flexibility and faster inference—have received limited attention in this field, and studies on Anchor-free frameworks capable of addressing the three core challenges above are even scarcer. This dual gap, in multi-challenge integration and Anchor-free development, motivates our proposed aspect-ratio-aware one-stage oriented ship detection method with dynamic label assignment.

3. Methodology

3.1. Overall framework

To address the core challenges of angle boundary discontinuity, the scarcity of high-quality features for high-aspect-ratio ships, and multi-scale variations in oriented ship detection in remote sensing images, we propose an integrated one-stage method with dynamic label assignment. The overall architecture of the proposed framework is illustrated in Fig. 2, where the modules work together to enhance feature extraction. Furthermore, a streamlined workflow that summarizes the complete training and inference process is provided in Fig. 3.

During inference, the input image first undergoes backbone-based feature extraction, with raw features sent to Feature Pyramid Networks (FPN) for multi-scale feature maps. These maps are then fed into our AGFFM to enhance and fuse scale-specific features, addressing multi-scale variations. The prediction head outputs centerness, classification, positional parameters, and angle encodings—where the latter are decoded by TNC to generate final OBBs, resolving angle boundary discontinuity.

During training, the RDDA module acts as a key link: it performs aspect-ratio-aware label assignment to increase high-quality positive samples for elongated ships. For these samples, we compute Gaussian centerness, labels, positional parameters, and angles—with angles encoded by TNC to ensure boundary continuity. The network is optimized via four loss functions, aligning training with inference demands.

3.2. Trigonometric normalized coder

To resolve the angle boundary discontinuity problem, we propose TNC, a concise angle encoder ensuring continuous representation near boundaries. Following the le90 long-edge definition, TNC creates equal encoded values at $-\pi/2$ and $\pi/2$ using trigonometric functions:

$$z = f(\theta) = [\sin 2\theta, \cos 2\theta] \quad (1)$$

where θ is the angle value in range $[-\pi/2, \pi/2]$, z represents the angle encoding vector, and $f(\cdot)$ is the encoding function. For decoding, TNC first normalizes the encoded vector and then applies the arctangent function to restore the original angle:

$$\theta = f^{-1}(z) = \frac{1}{2} \text{atan2} \left(\frac{z_1}{\sqrt{z_1^2 + z_2^2}}, \frac{z_2}{\sqrt{z_1^2 + z_2^2}} \right) \quad (2)$$

where z_1 and z_2 represent the first and second components of the angle encoding vector, respectively, atan2 is the four-quadrant arctangent function, which is used to determine the quadrant of the angle based on the signs of z_1 and z_2 , and f^{-1} represents the decoding function. The pipeline of the TNC is demonstrated in Fig. 2(a).

To enhance the TNC's robustness, we incorporate dual-frequency phase-shifting methodology inspired by Yu and Da (2023):

$$z = f_d(\theta) = [\sin 2\theta, \cos 2\theta, \sin 4\theta, \cos 4\theta] \quad (3)$$

where $f_d(\cdot)$ represents a dual-frequency encoding function that maps the angle into a 4-dimensional vector. The corresponding decoding function is as follows:

$$\theta = f_d^{-1}(\varphi_2, \delta) = \begin{cases} \frac{\pi + \varphi_2}{2}, & \text{if } \delta < 0 \\ \frac{\varphi_2}{2}, & \text{otherwise} \end{cases} \quad (4)$$

$$\delta = \cos(\varphi_1 - \varphi_2) \quad (5)$$

$$\varphi_1 = \text{atan2} \left(\frac{z_1}{\sqrt{z_1^2 + z_2^2}}, \frac{z_2}{\sqrt{z_1^2 + z_2^2}} \right) \quad (6)$$

$$\varphi_2 = \frac{1}{2} \text{atan2} \left(\frac{z_3}{\sqrt{z_3^2 + z_4^2}}, \frac{z_4}{\sqrt{z_3^2 + z_4^2}} \right) \quad (7)$$

where z_1 to z_4 represent the four components of the angular encoding vector, and $f_d^{-1}(\cdot)$ represents the dual-frequency decoding function. In summary, TNC enhances angle encoding accuracy and robustness by employing higher-dimensional vectors and multiple encoding frequencies, effectively creating a continuous representation space where boundary transitions become naturally smooth.

3.3. Ratio driven dynamic assignment

In remote sensing images, ships typically exhibit high aspect ratios, resulting in elongated bounding boxes with unevenly distributed feature quality. The quality of sample points varies significantly across different locations, with a limited number of high-quality sample points available. To address this limitation, we propose RDDA, which intelligently expands positive sample selection based on aspect ratios to incorporate these high-quality features that traditional approaches miss.

Our analysis reveals that under the same receptive field, features extracted from the ship's waist region are of higher quality than those from the head, as shown in Fig. 4. Conventional methods focusing only on points within the bounding box often overlook these valuable waist features. Since the varying aspect ratios of different ship targets lead to inconsistent feature quality distributions, dynamically adjusting the

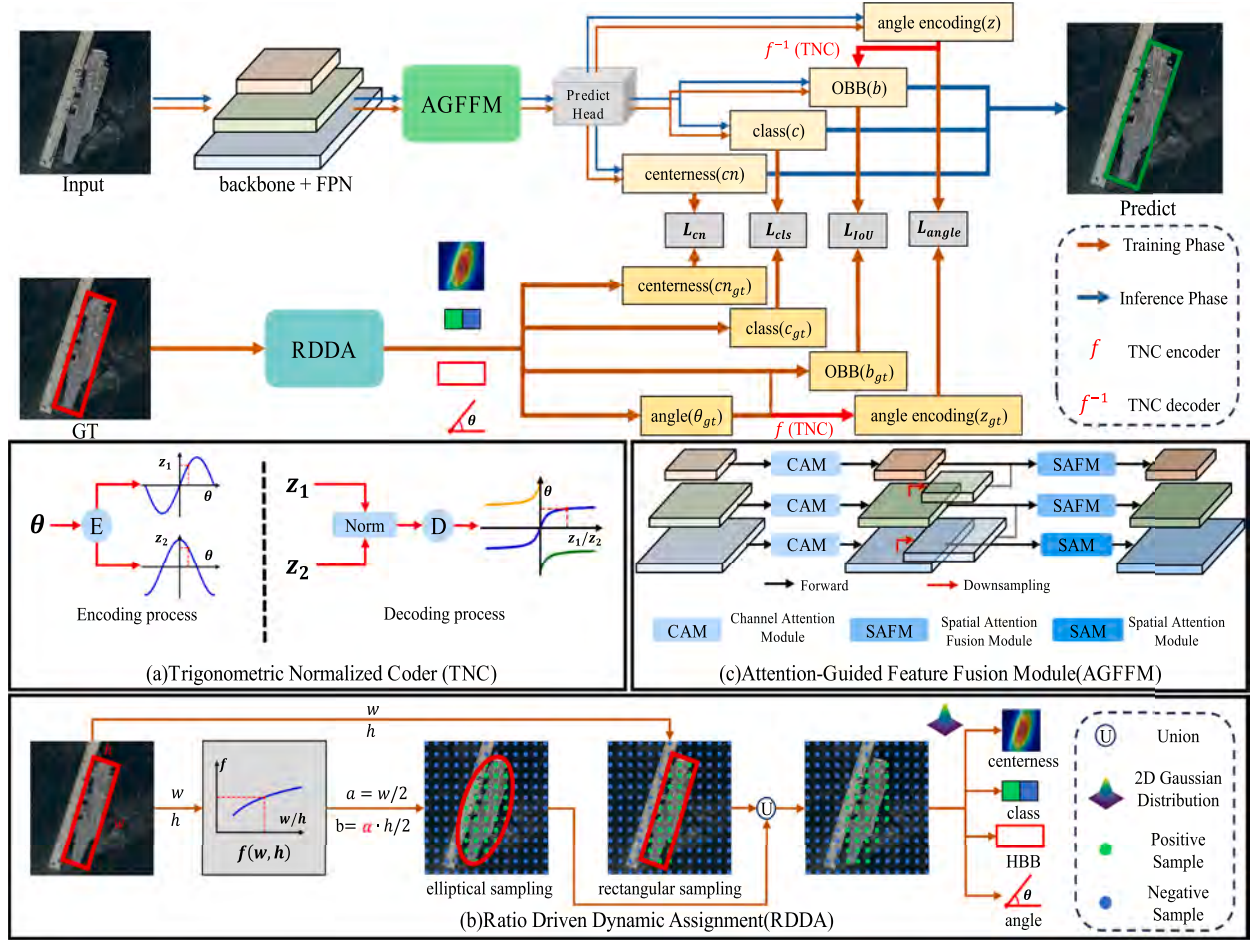


Fig. 2. The network structure of our ship detection framework. the upper part of the image shows the overall network architecture, and the lower part presents the detailed structures of the three modules TNC, RDDA and AGFFM. (a) the structure of trigonometric normalized coder (TNC). (b) the structure of ratio driven dynamic assignment (RDDA). (c) the structure of attention-guided feature fusion module (AGFFM).



Fig. 3. A simplified flowchart of the training and inference phases of our work. (a) Training Phase diagram (b) Inference Phase diagram.

sample point selection strategy based on the target's aspect ratio would be more conducive to improving detection accuracy.

As illustrated in Fig. 2(b), to leverage these high-quality waist features, we introduce elliptical sampling as a complement to traditional rectangular sampling. The sampling process initiates with standard rectangular sampling, assigning positive sample status to all feature map locations contained within ground truth bounding boxes. We then dynamically construct an ellipse based on the aspect ratio of ship to capture additional high-quality samples around the waist region. Sample points falling within the ellipse are regarded as positive samples.

Specifically, we define the shorter side of the GT box as height (h) and the longer side as width (w), thus creating an ellipse with semi-

minor axis ($b = \frac{h}{2} \cdot s$), semi-major axis ($a = \frac{w}{2}$), and center and orientation identical to the GT box. The short-axis scaling factor s adaptively expands based on aspect ratio:

$$s = \alpha \cdot \ln\left(\frac{w}{h}\right) + 1 \quad (8)$$

where α modulates the scaling effect of the short axis to control the expansion magnitude. The final positive samples are determined by the union of rectangular and elliptical sampling results.

Since RDDA includes points outside the GT box as positive samples, the original centerness calculation becomes ineffective. We address this by introducing Gaussian centerness to enable the assessment of feature



Fig. 4. Comparative analysis of receptive field efficiency in different ship regions. under the same receptive field, compared to sample points at the head of ship (midpoint of the bounding box's short side), those at the waist of ship (midpoint of the bounding box's long side) cover a broader portion of the hull. this results in the sample points located at the waist possessing a larger effective receptive field, thereby achieving higher quality. (a) head region receptive field. (b) waist region receptive field.

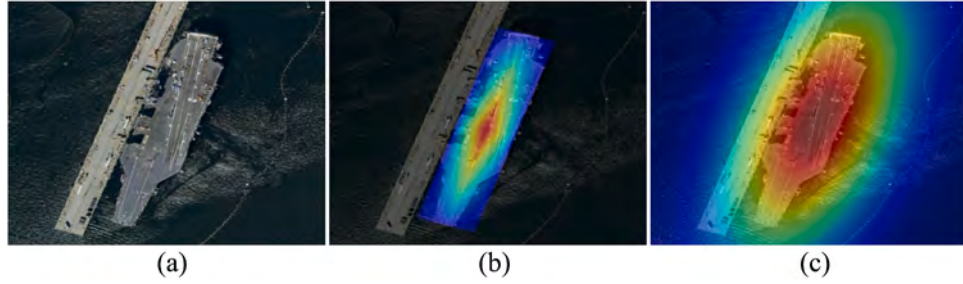


Fig. 5. Comparison between traditional and gaussian centerness measurements. (a) original image. (b) traditional centerness heatmap. (c) gaussian centerness heatmap.

quality for sample points outside the GT boxes. The ellipse is transformed into a two-dimensional Gaussian distribution:

$$\mu_0 = (x_0, y_0)^T \quad (9)$$

$$\Sigma^{1/2} = \mathbf{R}\mathbf{A}\mathbf{R}^T = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \quad (10)$$

where x_0 and y_0 represent the ellipse center, while θ denotes its rotation angle. For any sample point, its Gaussian centerness is calculated as:

$$\mathbf{x} = (x, y)^T \quad (11)$$

$$\text{centerness}_g(\mathbf{x}) = \exp\left(-\frac{1}{2}(\mathbf{x} - \mu)^T \Sigma^{-1}(\mathbf{x} - \mu)\right) \quad (12)$$

where x and y represent the coordinates of the sample point, $\text{centerness}_g(\cdot)$ denotes the Gaussian centerness function.

To illustrate the effectiveness of our approach, Fig. 5 compares both centerness measurement techniques applied to the same image. The visualization clearly shows that while traditional centerness only evaluates points within bounding boxes, our Gaussian centerness can effectively evaluate points both inside and outside the GT box. This expanded measurement capability allows our model to effectively incorporate high-quality waist features, significantly improving detection performance for ships with extreme aspect ratios.

3.4. Attention-guided feature fusion module

Due to the substantial scale differences among ships in remote sensing imagery, single-scale feature representations prove inadequate for reliable detection. While FPN provide multi-scale representation, their

simple top-down fusion approach often fails to fully exploit complementary information across scales. To address this limitation, we propose AGFFM, which integrates both channel and spatial attention mechanisms to enhance feature representation, as illustrated in Fig. 2(c).

Our module first applies channel attention to emphasize informative feature channels:

$$F_c^i = F^i \odot \text{CAM}(F^i) \quad (13)$$

where F^i denotes the i -th layer feature extracted by FPN, with $i \in [1, n]$, $\text{CAM}(\cdot)$ represents the Channel Attention Module introduced in Woo et al. (2018), and F_c^i denotes the feature after channel attention processing.

Subsequently, downsampling is performed on features within the range $i \in [1, n-1]$ to obtain F_{cd}^i . Thereafter, feature fusion is performed between each layer's feature and the downsampled feature of its adjacent layer:

$$F_f^i = \text{SAFM}(F_{cd}^{i-1}, F_c^i) \quad (14)$$

where F_f^i is the fused feature and $i \in [2, n]$. Inspired by Li et al. (2023), we employ the spatial attention mechanism, called Spatial Attention Fusion Module(SAFM), to fuse features from two different sources. The specific fusion process will be elaborated in the subsequent sections.

Then, perform feature enhancement on F_c^1 individually along the spatial dimension:

$$F_{cs}^1 = F_c^1 \odot \text{SAM}(F_c^1) \quad (15)$$

where F_{cs}^1 represents the features of F_c^1 after spatial attention, and $\text{SAM}(\cdot)$ denotes the Spatial Attention Module as described in Woo et al. (2018). Finally, we utilize all the fused features F_f^i along with F_{cs}^1 as the final multi-scale features, which are then passed to the subsequent classification head and regression head for classification and regression tasks.

For the feature fusion process $F_f = \text{SAFM}(F_1, F_2)$ mentioned above, the specific approach is as follows. We first perform channel-wise concatenation on the two features F_1 and F_2 to obtain F_{con} . Then, channel-wise average pooling and max pooling operations are applied to aggregate spatial information from the feature maps, yielding SF_{avg} and SF_{max} respectively. Following this, these spatial features are concatenated, convolved, and activated through a sigmoid function to obtain the spatial attention maps:

$$[SA_1, SA_2] = \text{Sigmoid}(\text{Conv}(\text{Concat}(SF_{avg}, SF_{max}))) \quad (16)$$

where $\text{Conv}(\cdot)$ denotes a convolution operation with both input and output channels set to 2, SA_1 and SA_2 represent the features on channel 1 and channel 2 after convolution. The final fused feature F_f is obtained through:

$$F_f = SA_1 \odot F_1 + SA_2 \odot F_2 \quad (17)$$

By integrating attention mechanisms at multiple levels, AGFFM effectively enhances scale-specific information while preserving cross-scale complementarity, significantly improving the model's ability to detect ships of various scales.

3.5. Loss function

The overall loss function comprises four distinct components: classification loss, IoU loss, centerness loss, and angle encoding loss, which can be formulated as follows:

$$L = \alpha L_{cls}(c, c_{gt}) + \beta L_{IoU}(b, b_{gt}) + \gamma L_{cn}(cn, cn_{gt}) + \lambda L_{angle}(z, z_{gt}) \quad (18)$$

where c , b , cn , and z denote the predicted class, bounding box, centerness, and angle encoding, with corresponding ground truth values c_{gt} , b_{gt} , cn_{gt} , and z_{gt} . $L_{cls}(\cdot)$ is computed using the focal loss, $L_{IoU}(\cdot)$ employs the rotated IoU loss, $L_{cn}(\cdot)$ utilizes the cross-entropy loss, and $L_{angle}(\cdot)$ is calculated using the L1 loss. α , β , γ and λ are weight parameters.

4. Experiments and analysis

In this section, we evaluate the effectiveness of our proposed method on three public datasets for oriented ship detection. We first describe the datasets and implementation details, then present contrast experiments with state-of-the-art methods, comprehensive ablation studies, and parameter experiments.

4.1. Datasets

4.1.1. HRSC2016 (Liu et al., 2017)

This high-resolution remote sensing dataset for ship detection from Northwestern Polytechnical University contains 1061 images with 2970 ship instances annotated with OBBs. The sizes of image range from 700 to 1100 pixels. Following the official split, we use 617 images for training and 444 for testing. All images are resized to 800×512 pixels and augmented with random flipping and rotation to enhance the robustness of our model during training.

4.1.2. HRSID (Yue et al., 2023)

Released by the Institute of Electronics, Chinese Academy of Sciences, this large-scale SAR imagery dataset features 5604 images 800×800 with 16951 ship instances annotated with OBBs, including challenging scenarios like near-shore ships, adjacent ships, and densely packed small ships. Images have resolutions ranging from 1 to 5 meters. Following the official partition, we use a dataset of 3642 images for training and a separate set of 1962 images for testing, with random flipping augmentation.

Table 2

Experimental hyperparameters.

Hyperparameter Type	Parameter	Value
Training Epochs	Epochs on HRSC2016	72
	Epochs on HRSID	100
	Epochs on SSDD +	100
Optimizer Parameters	Optimizer	AdamW
	Learning Rate	1×10^{-4}
	β_1	0.9
	β_2	0.999
	Weight Decay	0.05
Training Settings	Batch Size	2
Parameters in Eq. (18)	α	1
	β	1
	γ	1
	λ	0.1
Parameter α in Eq. (8)	α on HRSC2016	0.3
	α on HRSID	0.4
	α on SSDD +	0.3

4.1.3. SSDD + (Zhang et al., 2021b)

We utilize an augmented version of the SSDD dataset, where 1,160 SAR ship images have been annotated with OBBs. The multi-sensor dataset includes imagery from Radarsat-2, TerraSAR-X, and Sentinel-1 satellites, with spatial resolutions varying between 1-15 meters. We follow the official 8:2 train-test split and resize all images to 608×608 pixels, applying random flipping and rotation augmentations.

4.2. Implementation details

We conduct experiments on all three datasets. To ensure the fairness of the experiment, we apply a batch size of 2 for all compared methods on the same dataset. All models are trained for 72 epochs on HRSC2016, 100 epochs on HRSID, and 100 epochs on SSDD +. For all experiments, we employ the AdamW optimizer with a learning rate of 0.0001, β_1 set to 0.9, β_2 to 0.999, and a weight decay of 0.05. The parameters α , β , γ and λ in Eq. (18) are set to 1, 1, 1 and 0.1, respectively. In Eq. (8), the hyperparameter α is set to 0.3 for the HRSC2016 and SSDD + datasets, and 0.4 for the HRSID dataset. The corresponding parameter sensitivity experiments are presented in Section 4.5. For clarity, all hyperparameters used in this study are summarized in Table 2.

All experiments are performed on an NVIDIA GeForce RTX 4090 GPU with Ubuntu 22.04.2 LTS, using PyTorch 2.0.0 and MMrotate 1.0.0rc1 (Zhou et al., 2022). For evaluation, we employ the PASCAL VOC metric on HRSC2016 and SSDD + according to their data formats, and the MS COCO metrics on HRSID for multi-scale targets. In the PASCAL VOC metric, mAP represents the mean average precision across all categories at IoU = 0.5. In the MS COCO metric, AP_{50} and AP_{75} represent the AP at IoU thresholds of 0.5 and 0.75, respectively. $AP_{50:95}$ donates the average AP over IoU thresholds ranging from 0.5 to 0.95. AP_S , AP_M , and AP_L represent the AP values of the model for small targets (with area under 32×32 pixels), medium targets (with area between 32×32 and 96×96 pixels), and large targets (with area over 96×96 pixels), respectively.

4.3. Contrast experiments

To demonstrate our method's effectiveness, we compare it with several state-of-the-art methods across all three datasets. Comparison methods include one-stage algorithms (KFIOU Yang et al., 2022, R³Det Yang et al., 2021a, KLD Yang et al., 2021c, FPNFormer Tian et al., 2024, ATSS Zhang et al., 2020, GWD Yang et al., 2021b, PSC Yu & Da, 2023, S²ANet Han et al., 2021a and ACM Xu et al., 2024), and two-stage algorithms (ReDet Han et al., 2021b, RoI-Transformer Ding et al., 2019). All implementations use the mmrotate framework (Zhou et al., 2022) with consistent data augmentation and training strategies.

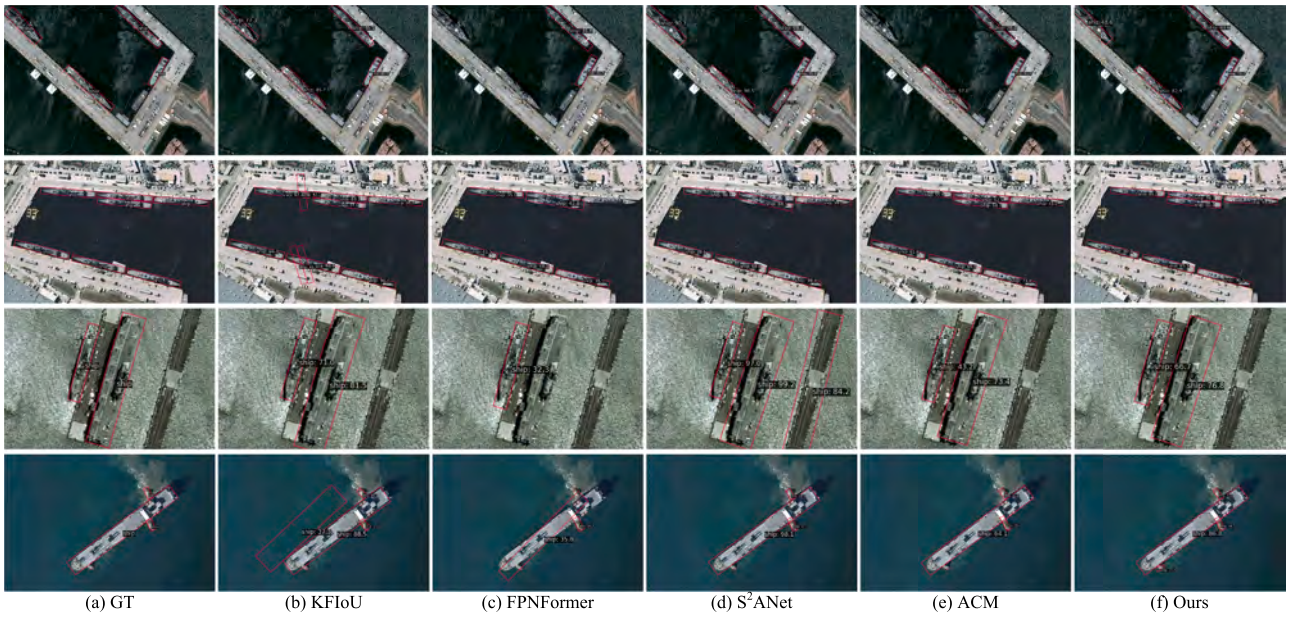


Fig. 6. Prediction results of different models on HRSC2016. (a) ground truth. (b) KFIoU. (c) FPNFormer. (d) S²ANet. (e) ACM. (f) Ours.

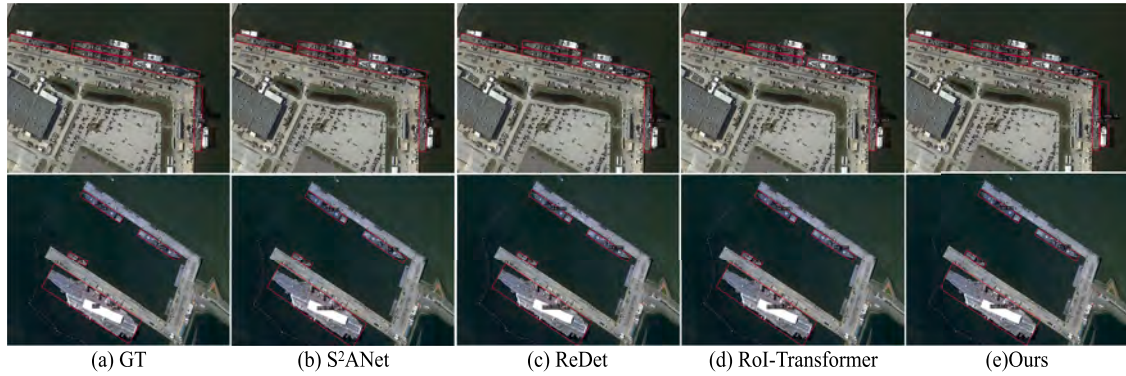


Fig. 7. Visualization of comparative results where the proposed method does not perform optimally on HRSC2016. (a) Ground truth. (b) S²ANet (c) ReDet. (d) RoI-Transformer. (e) Ours.

Table 3
Comparison with state-of-the-art methods on HRSC2016.

Type	Method	Base Model	FPS	mAP(%)
Two-Stage	ReDet	Cascade RCNN	6.8	90.66
	RoI-Transformer	Cascade RCNN	8.0	90.64
One-Stage	KFIoU	R ³ Det	28.3	90.47
	R ³ Det	R ³ Det	28.9	90.00
	KLD	R ³ Det	29.8	90.49
	FPNFormer	RetinaNet	5.3	77.67
	ATSS	ATSS	37.3	89.86
	GWD	RetinaNet	39.5	84.70
	PSC	RetinaNet	37.4	84.52
	S ² ANet	S ² ANet	28.0	90.52
One-Stage	ACM	FCOS	39.4	90.24
	Ours	FCOS	30.1	90.65

4.3.1. Results on HRSC2016

As shown in Table 3, our method achieves 90.65% mAP on HRSC2016, outperforming all one-stage detectors including S²ANet (+0.13%) and KLD (+0.16%). Our performance matches two-stage methods like RoI-Transformer (90.64%) and ReDet (90.66%) while offering significantly faster inference—30.1 FPS compared to 8.0 and 6.8 FPS respectively—providing an excellent accuracy-speed trade-off.

The qualitative results in Fig. 6 reveal our method's superior performance in challenging scenarios while other methods exhibit varying degrees of false positives and missed detections. For instance, S²ANet incorrectly identifies a floating platform in the first row and harbor structures in the third row as ships, and misses detections in densely arranged ships in the second row, while our approach still maintains high accuracy. Notably, our method successfully detects an unlabeled small ship in the fourth row that even the ground truth fails to label, demonstrating its robustness in challenging scenarios. This enhanced performance stems from our complementary modules: RDDA increases high-quality positive samples while AGFFM enables effective multi-scale feature enhancement. Additionally, Fig. 7 presents a comparison between our method and existing algorithms, including cases where our method does not perform optimally. It is worth noting that these limitations are rare and do not change the fact that our method achieves superior overall detection accuracy, comparable to state-of-the-art two-stage detectors. Specifically, our method detects targets effectively but yields lower confidence scores than S²ANet, ReDet and RoI-Transformer in some samples. In the first row of Fig. 7, it misidentifies shadows as ships in dense scenarios, revealing room for robustness improvement. These issues have limited influence on the overall accuracy and efficiency of our method. Analysis attributes low confidence to the one-stage architecture's lack of iterative feature refinement, and poor dense-scenario performance to the anchor-free design's insufficient feature constraints

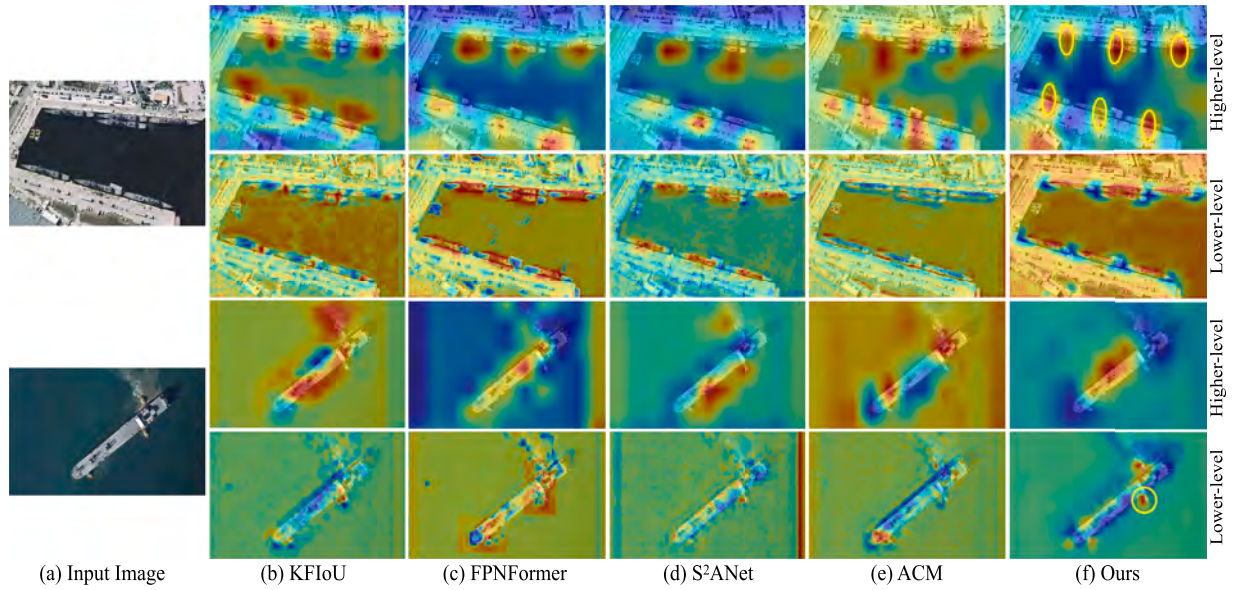


Fig. 8. Feature visualization on HRSC2016. the first and third rows represent the higher-level feature maps of the model, while the second and fourth rows display its lower-level feature maps. (a) Input image. (b) KFIoU. (c) FPNFormer. (d) s^2 anet. (e) ACM. (f) Ours.

and the framework's absence of regional screening. Based on our analysis, the low confidence stems from the algorithm's adoption of a one-stage architecture that lacks iterative regional feature refinement. Its poor performance in dense scenarios is due to the absence of explicit feature constraints in the anchor-free design, which easily leads to feature confusion between adjacent objects. Meanwhile, the lack of regional screening in the one-stage framework further exacerbates the false detection problem.

Additionally, as illustrated in Fig. 8, by utilizing the channel averaging and normalization strategy, we visualize the 2nd and 4th layers of the FPN output as the lower-level and higher-level feature maps, respectively, in which red regions represent areas of high attention and blue regions denote areas of low attention. It further demonstrates our method's superior feature extraction ability. As shown in the first row (indicated by the yellow circle), in higher-level feature maps, our approach produces more precise attention regions centered on ships. The enhanced focus on ship mid-body regions confirms RDDA's effectiveness in utilizing high-quality waist features. As shown in the fourth row (indicated by the yellow circle), in lower-level feature maps, our method more clearly detects small ships and extracts precise features, validating AGFFM's capability to capture multi-scale information.

4.3.2. Results on HRSID

Table 4 shows our method's strong performance on HRSID. Compared to the best one-stage competitor (S^2 ANet), we achieve substantial improvements across all metrics: +3.1% in $AP_{50:90}$, +0.4% in AP_{50} , +5.1% in AP_{75} , +1.9% in AP_S , +5.8% in AP_M , and a remarkable +16.6% in AP_L . While two-stage methods slightly outperform ours in some metrics, our approach shows advantages in AP_{50} (+0.7% over RoI-Transformer, +0.9% over ReDet) and excels in large object detection (+4.3% over RoI-Transformer, +2.9% over ReDet in AP_L).

Fig. 9 presents qualitative comparisons on HRSID. It is evident that our method achieves superior performance. In challenging scenes with shore-side structures, all competing methods incorrectly identify port infrastructure as ships (indicated by yellow boxes) shown in the second row, while our method produces results consistent with ground truth. This superior discrimination capability stems from AGFFM's ability to extract comprehensive multi-scale features that include contextual background information, preventing the model from being misled by local similarities and enhancing robustness against complex maritime environments.

4.3.3. Results on SSDD +

The contrast experimental results on SSDD + dataset are shown in Table 5. Our method achieves the best mAP performance (92.60%), outperforming all other one-stage detectors including S^2 ANet (+0.11%), KFIoU (+0.75%), and ACM (+1.47%). While two-stage detectors like ReDet and RoI-Transformer achieve slightly higher accuracy, our method operates at 31.4 FPS—nearly 5 times faster than their 6.7 FPS.

Fig. 10 shows our method's advantages in challenging scenarios. In nearshore environments (first row), competitors like PSC and ACM fail to detect large coastal ships due to shoreline interference. Our method not only successfully identifies these ships but also demonstrates more precise angle prediction compared to S^2 ANet, validating TNC's effectiveness in resolving angular boundary discontinuity and confirming our approach's robustness for SAR-based maritime surveillance.

4.4. Ablation study

To validate the effectiveness of our proposed components, we conducted comprehensive ablation experiments on all three datasets. The results are shown in Tables 6 to 8. Here, we use an FCOS (Tian et al., 2019) model with ConvNeXt (Liu et al., 2022) backbone as our baseline, where angles follow the le90 convention.

4.4.1. Effectiveness of the TNC

The proposed TNC aims to mitigate the issue of angle discontinuity in remote sensing-oriented ship detection. On the HRSC2016 dataset (Table 6), TNC alone improves mAP by 0.10% over the baseline. On the HRSID dataset (Table 7), TNC strategy enhances all metrics ($AP_{50:90}$, AP_{50} , AP_{75} , AP_S , AP_M , and AP_L), with a particularly notable 11.3% gain in AP_L , which effectively demonstrates the good ability of proposed angle encoding to mitigate angle discontinuity. Similarly, on SSDD + (Table 8) TNC strategy improves the baseline by 0.26%.

The qualitative analysis in Fig. 11 further illustrates TNC's benefits. While the baseline exhibits reasonable bounding box size regression, it struggles with angle prediction, particularly evident in the third row where angle errors lead to increased IoU between detection boxes of densely packed ships, causing missed detections after non-maximum suppression. TNC effectively resolves these issues through its continuous angle representation, significantly improving angle regression accuracy.

Table 4
Comparison with state-of-the-art methods on HRSID.

Type	Method	Angle	Params(<i>M</i>)	FPS	$AP_{50:90}(\%)$	$AP_{50}(\%)$	$AP_{75}(\%)$	$AP_S(\%)$	$AP_M(\%)$	$AP_L(\%)$
Two-Stage	ReDet	le90	31.36	4.2	56.6	85.7	64.4	55.0	64.7	35.0
	RoI-Transformer	le90	55.05	5.3	57.1	85.9	63.7	55.2	65.6	33.6
One-Stage	KFIoU	oc	41.58	24.1	48.5	84.9	51.3	46.6	57.0	17.2
	R ³ Det	oc	41.58	25.7	45.5	83.0	46.9	43.8	54.0	12.3
	KLD	oc	36.86	27.4	46.7	83.7	48.4	45.3	55.2	12.2
	FPNFormer	le90	48.77	3.8	41.1	75.7	41.3	40.8	46.1	3.0
	ATSS	le90	36.02	30.8	48.6	83.5	51.7	46.8	58.2	17.3
	GWD	le90	36.13	32.5	46.6	80.5	48.6	46.7	50.0	13.8
	PSC	le90	36.19	31.4	43.4	78.8	43.8	41.5	52.7	11.0
	S ² ANet	le90	36.18	24.6	50.2	86.2	53.5	49.2	57.1	21.3
	ACM	le90	31.90	33.2	49.8	83.5	53.5	48.1	59.2	23.8
One-Stage	Ours	le90	36.18	27.9	53.3	86.6	58.6	51.1	62.9	37.9

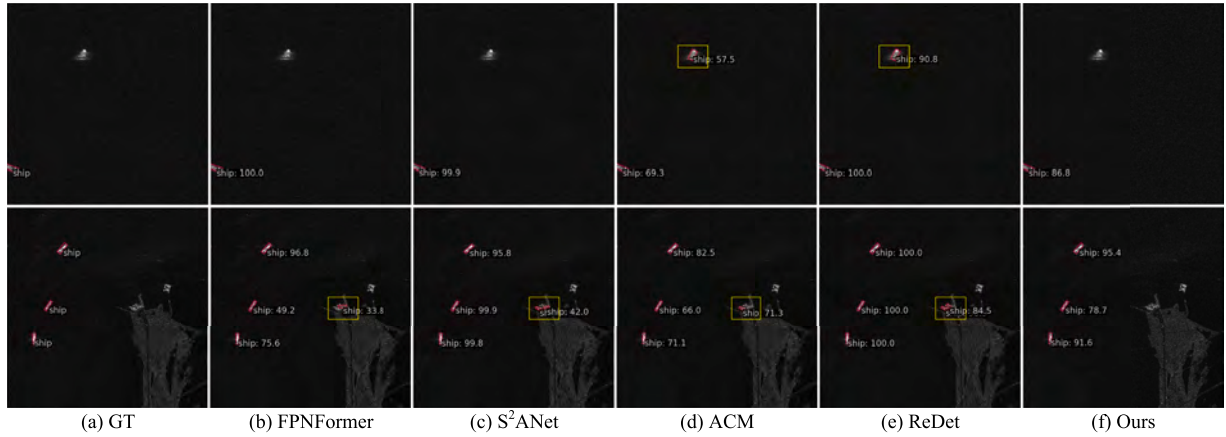


Fig. 9. Prediction results of different models on HRSID. (a) Ground truth. (b) FPNFormer. (c) S²ANet. (d) ACM. (e) ReDet. (f) Ours.

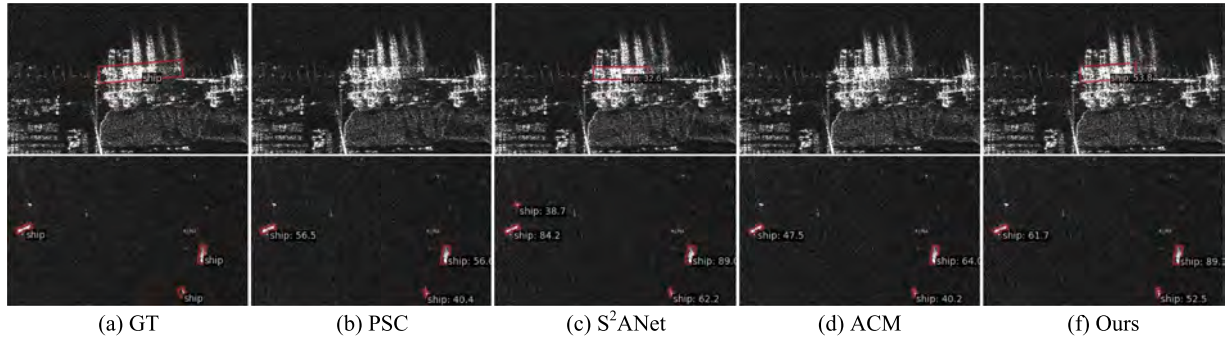


Fig. 10. Prediction results of different models on SSDD+. (a) Ground truth. (b) PSC. (c) S²ANet. (d) ACM. (e) Ours.

Table 5
Comparison with state-of-the-art methods on SSDD+.

Type	Method	FPS	<i>mAP</i> (%)
Two-Stage	ReDet	6.7	94.10
	RoI-Transformer	6.7	93.15
One-Stage	KFIoU	26.1	91.85
	R ³ Det	26.8	88.64
	KLD	29.2	90.01
	FPNFormer	5.6	86.72
	ATSS	32.1	90.93
	GWD	36.9	89.15
	PSC	34.4	88.80
	S ² ANet	25.8	92.49
	ACM	35.2	91.13
One-Stage	Ours	31.4	92.60

Table 6
Ablation experiments on HRSC2016.

Method	TNC	RDDA	AGFFM	<i>mAP</i> (%)
Baseline	✗	✗	✗	90.28
Ours	✓	✗	✗	90.38(+0.10)
	✓	✓	✗	90.56(+0.28)
	✓	✓	✓	90.65(+0.37)

4.4.2. Effectiveness of the RDDA

Our RDDA module addresses the challenges posed by high aspect ratio ships by dynamically expanding positive sample selection based on the target's aspect ratio to increase the number of positive samples. On HRSC2016 (Table 6), adding RDDA to TNC increases *mAP* by an additional 0.18% (0.28% total improvement over baseline). On HRSID, this combination yields substantial gains of 1.7% in $AP_{50:90}$, 1% in AP_{50} ,

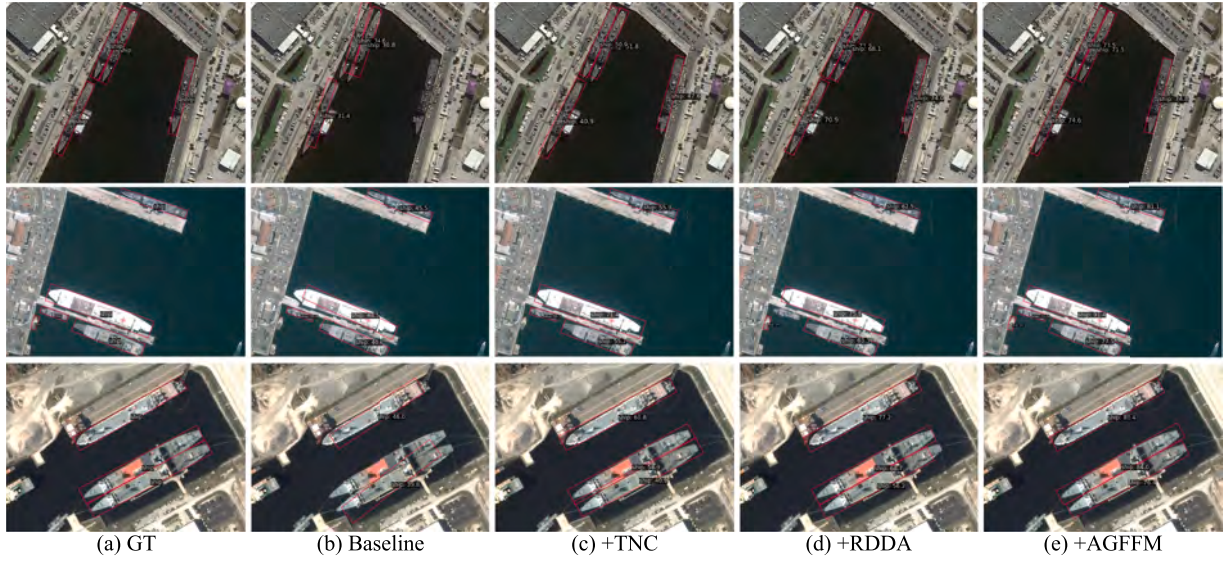


Fig. 11. Visual detection results of baseline method and our proposed strategies on HRSC2016. (a) Ground truth. (b) Baseline. (c) baseline with TNC. (d) Baseline with TNC and RDDA. (e) baseline with TNC, RDDA and AGFFM.

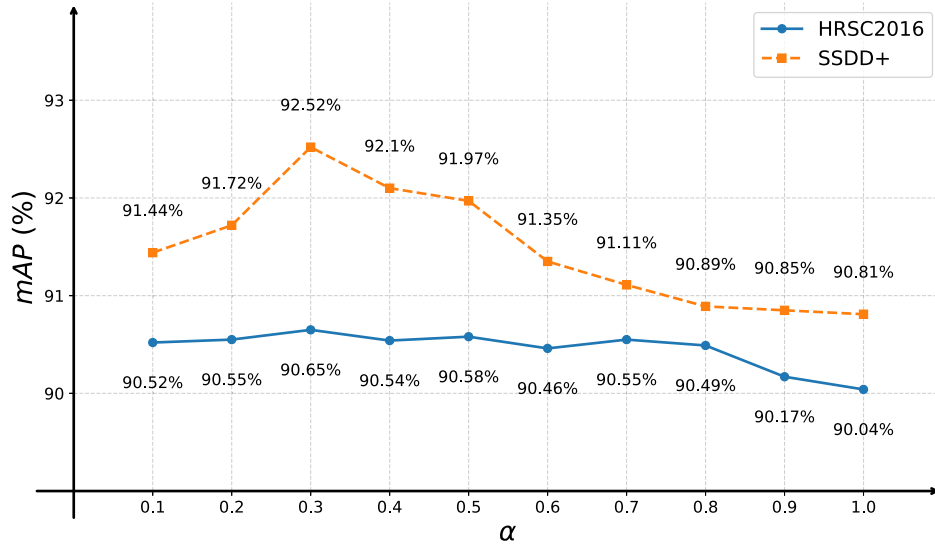


Fig. 12. The parameter experiments for α on HRSC2016 and SSDD+.

Table 7

Ablation experiments on HRSID.

Method	TNC	RDDA	AGFFM	$AP_{50:90}(\%)$	$AP_{50}(\%)$	$AP_{75}(\%)$	$AP_S(\%)$	$AP_M(\%)$	$AP_L(\%)$
Baseline	✗	✗	✗	48.0	84.2	50.8	46.2	56.8	24.5
Ours	✓	✗	✗	50.8(+2.8)	84.8(+0.6)	55.4(+4.6)	48.2(+2.0)	61.9(+5.1)	35.8(+11.3)
	✓	✓	✗	52.5(+4.5)	85.8(+1.6)	56.8(+6.0)	50.1(+3.9)	62.8(+6.0)	35.8(+11.3)
	✓	✓	✓	53.3(+5.3)	86.6(+2.4)	58.6(+7.8)	51.1(+4.9)	62.9(+6.1)	37.9(+13.4)

Table 8

Ablation experiments on SSDD+.

Method	TNC	RDDA	AGFFM	$mAP(\%)$
Baseline	✗	✗	✗	91.37
Ours	✓	✗	✗	91.63(+0.26)
	✓	✓	✗	92.10(+0.73)
	✓	✓	✓	92.60(+1.23)

and 1.4% in AP_{75} in Table 7. For SSDD+, integrating RDDA with TNC enhances mAP by 0.47% over TNC alone. All the results effectively demonstrate that the additional high-quality positive samples provided by RDDA benefit target parameter regression.

Fig. 11 demonstrates that combining RDDA with TNC significantly increases the model's confidence scores for ship targets, indicating enhanced detection certainty. This improvement stems from RDDA's ability to select higher-quality sample points during training, validating

Table 9

Comparative experiments of different attention modules on HRSC2016.

Method	mAP(%)
Baseline	90.56
SE	90.49
CBAM	90.58
ECA	90.42
AGFFM	90.65

our dynamic assignment strategy's effectiveness in optimizing feature utilization.

4.4.3. Effectiveness of the AGFFM

Our AGFFM module enhances multi-scale feature representation through attention-guided fusion. On HRSC2016, adding AGFFM provides a further 0.09% mAP improvement from 90.56% to 90.65% (Table 6). On HRSID, AGFFM enhances all metrics, with notable improvements of 1.0%, 0.1%, and 2.1% in AP_S , AP_M , and AP_L respectively, demonstrating its effectiveness across objects of varying scales as shown in Table 7. On SSDD+, AGFFM contributes a 0.50% improvement, increasing mAP from 92.10% to 92.60% in Table 8.

To further demonstrate the effectiveness of the AGFFM module, we conduct comparative experiments between AGFFM and several commonly used attention modules, including SE attention (Hu et al., 2018), CBAM attention (Woo et al., 2018), and ECA attention (Wang et al., 2020). The experimental results are presented in Table 9. The baseline incorporates the TNC and RDDA strategies. It can be observed that the AGFFM proposed in our method outperforms other attention mechanisms on the HRSC2016 dataset, achieving a 0.07% improvement compared with the top-performing CBAM algorithm.

The qualitative results in Fig. 11 show that AGFFM substantially improves detection performance, particularly for small objects. In the second row, the model without AGFFM exhibits limited multi-scale detection capability and poor performance on small objects, while our complete model successfully identifies all ships, including smaller ships. This confirms AGFFM's effectiveness in enhancing multi-scale feature representation.

4.5. Parameter experiments

To determine the optimal value for hyperparameter α in Eq. (8), which controls the expansion of the elliptical sampling region in our RDDA module, we conducted systematic experiments across all three datasets. We maintained all other settings while varying α from 0.1 to 1.0 in increments of 0.1. As α increases, the short axis of the elliptical sampling region expands, incorporating more points as positive samples. Results shown in Table 10 and Fig. 12 reveal consistent patterns across datasets.

On HRSC2016, performance initially improves with increasing α , reaching peak mAP at $\alpha = 0.3$, before declining at higher values. This pattern demonstrates a clear trade-off: when α is too small, valuable features near the waist of ship are overlooked, limiting RDDA's effectiveness; conversely, when α is too large, low-quality background points contaminate the positive sample set, degrading detection accuracy. Similar trends appear on other datasets. On HRSID, optimal performance occurs at $\alpha = 0.4$, where multiple metrics reach their peak simultaneously— $AP_{50:90}$, AP_{75} and AP_S achieve maximum values, while AP_{50} and AP_M rank second and third highest respectively. This convergence of optimal metrics indicates that $\alpha = 0.4$ provides the best overall detection performance on this dataset. As for SSDD+, mAP peaks at $\alpha = 0.3$, matching the optimal value found for HRSC2016.

Table 10

Parameter experiments for α on HRSID. the top three within each metric are highlighted in bold, italic and underline.

α	$AP_{50:90}$ (%)	AP_{50} (%)	AP_{75} (%)	AP_S (%)	AP_M (%)	AP_L (%)
0.1	52.6	85.4	57.3	50.3	62.6	39.1
0.2	53.2	86.4	<u>57.9</u>	<u>50.8</u>	63.2	<u>38.6</u>
0.3	<i>53.1</i>	86.5	<i>58.3</i>	50.7	<i>63.0</i>	38.7
0.4	53.3	<i>86.6</i>	58.6	51.1	<u>62.9</u>	37.9
0.5	52.9	85.7	57.6	50.8	62.7	38.1
0.6	<u>52.9</u>	86.9	57.4	<i>50.9</i>	62.0	34.2
0.7	52.5	86.0	<u>57.9</u>	50.2	62.6	32.9
0.8	52.4	86.9	56.7	50.3	62.0	32.7
0.9	52.2	86.0	56.6	50.0	61.7	31.1
1.0	52.5	86.0	56.6	50.1	62.2	33.5

These consistent findings across datasets suggest that moderate expansion of the sampling region effectively balances the inclusion of high-quality waist features against the risk of incorporating background noise, confirming the effectiveness of our aspect-ratio-aware sampling strategy while providing practical guidance for implementation.

5. Conclusion

In this paper, we propose a one-stage anchor-free rotated ship detector with three targeted innovative components, addressing the key challenges of angle boundary discontinuity, high-aspect-ratio feature scarcity, and insufficient multi-scale representation in remote sensing ship detection. Our core contributions lie in three novel modules: the Trigonometric Normalized Coder (TNC) uses simple trigonometric functions to build a continuous angle space, fundamentally resolving the inherent boundary discontinuity issue of traditional rotated detection methods with high computational efficiency; the Ratio Driven Dynamic Assignment (RDDA) uniquely leverages ship aspect ratios to expand positive sample regions to the overlooked waist features of elongated ships, effectively alleviating high-quality sample scarcity for slender targets; the Attention-Guided Feature Fusion Module (AGFFM) enhances multi-scale features via dual attention mechanisms, adaptively highlighting ship features and suppressing background interference. Validated on three benchmark datasets, our method achieves superior detection accuracy, for example 90.65% mAP on HRSC2016, while maintaining efficient inference speed. It provides a practical solution for remote sensing maritime surveillance and advances the application of one-stage anchor-free detectors in high-aspect-ratio oriented object detection tasks.

We also note several limitations. In a small subset of samples, the confidence scores of detections from our one-stage framework are slightly lower than those of two-stage networks, as the latter employs iterative region feature refinement to boost target recognition accuracy. Future work will focus on narrowing this minor gap and extending the model to detect other high-aspect-ratio objects analogous to ships, such as trucks and bridges in remote sensing imagery.

CRedit authorship contribution statement

Dandan Ma: Conceptualization, Methodology, Formal analysis, Writing – original draft, Writing – review & editing, Visualization, Supervision; **Hao Huang:** Methodology, Investigation, Data curation, Validation, Visualization, Writing – original draft; **Zhiyu Jiang:** Conceptualization, Formal analysis, Resources, Visualization, Funding acquisition, Supervision, Writing – review & editing.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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